

Probabilistic Robotics

Autonomous Navigation of Mobile Robots

Anna Semakova

a.semakova@gmail.com

06.03.2025

Key Idea

Explicit representation of uncertainty
(using the calculus of probability theory)

- Perception = state estimation
- Action = utility optimization

Axioms of Probability Theory

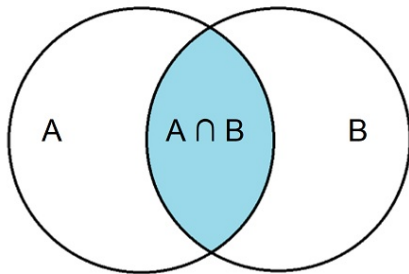
$P(A)$ denotes probability that proposition A is True

- $0 \leq P(A) \leq 1$
- $P(\text{True}) = 1, \quad P(\text{False}) = 0$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Axiom 3

a Closer Look

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



Using the Axioms

$$P(A \cup \neg A) = P(A) + P(\neg A) - P(A \cap \neg A)$$

$$P(\text{True}) = P(A) + P(\neg A) - P(\text{False})$$

$$1 = P(A) + P(\neg A) - 0$$

$$P(\neg A) = 1 - P(A)$$

Random Variables

Discrete

- X denotes a **random variable**
- X can take on a countable number of values in $\{x_1, x_2, \dots, x_n\}$
- $P(X = x_i)$ or $P(x_i)$ is the **probability** that the random variable X takes on value x_i
- $P(\cdot)$ is called **probability mass function**

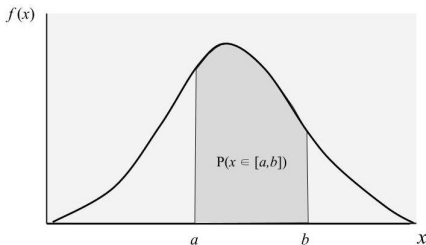
- $P(\text{Room}) = \langle 0.7, 0.2, 0.08, 0.02 \rangle$
Room $\in \{\text{office, corridor, lab, kitchen}\}$

Random Variables

Continuous

- X takes on values in the continuum
- $p(X = x)$ or $p(x)$ is a **probability density function**

$$P(x \in [a, b]) = \int_a^b p(x) dx$$



Probability Sums up to One

- Discrete case

$$\sum_x P(x) = 1$$

- Continuous case

$$\int p(x) dx = 1$$

Joint and Conditional Probability

- $P(X = x \text{ and } Y = y) = P(x, y)$
- If X and Y are independent then

$$P(x, y) = P(x) \cdot P(y)$$

- $P(x | y)$ is the probability of x given y

$$P(x | y) = P(x, y) / P(y)$$

$$P(x, y) = P(x | y) \cdot P(y)$$

- If X and Y are independent then

$$P(x | y) = P(x)$$

Law of Total Probability

■ Discrete case

$$P(x) = \sum_y P(x|y) \cdot P(y)$$

■ Continuous case

$$p(x) = \int p(x|y) \cdot p(y) dy$$

Marginalization

- Discrete case

$$P(x) = \sum_y P(x, y)$$

- Continuous case

$$p(x) = \int p(x, y) dy$$

Bayes Formula

$$P(x, y) = P(x | y) \cdot P(y) = P(y | x) \cdot P(x) \Rightarrow$$

$$P(x | y) = \frac{P(y | x) \cdot P(x)}{P(y)} = \frac{\text{likelihood} \cdot \text{prior}}{\text{evidence}}$$

Normalization

$$P(x|y) = \frac{P(y|x) \cdot P(x)}{P(y)} = \eta \cdot P(y|x) \cdot P(x)$$

$$\eta = P(y)^{-1} = \frac{1}{\sum_x P(y|x) \cdot P(x)}$$

Algorithm:

$$\forall x: \quad \text{aux}_{x|y} = P(y|x) \cdot P(x)$$

$$\eta = \frac{1}{\sum_x \text{aux}_{x|y}}$$

$$\forall x: \quad P(x|y) = \eta \cdot \text{aux}_{x|y}$$

Bayes Rule

with Background Knowledge

$$P(x | y, z) = \frac{P(y | x, z) \cdot P(x | z)}{P(y | z)}$$

Conditional Independence

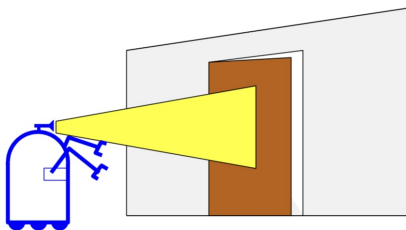
$$P(x, y | z) = P(x | z) \cdot P(y | z)$$

- Equivalent to $P(x | z) = P(x | z, y)$ and $P(y | z) = P(y | z, x)$
- But this does not necessarily mean $P(x, y) = P(x) \cdot P(y)$

State Estimation

Simple Example

- Suppose a robot obtains measurement z
- What is $P(\text{open} \mid z)$?



Causal vs. Diagnostic Reasoning

- $P(\text{open} | z)$ is **diagnostic**
- $P(z | \text{open})$ is **causal** (count frequencies)
- In some situations, **causal** knowledge is easier to obtain
- Bayes rule allows us to use causal knowledge:

$$P(\text{open} | z) = \frac{P(z | \text{open}) \cdot P(\text{open})}{P(z)}$$

Example

- $P(z \mid \text{open}) = 0.6$ $P(z \mid \neg \text{open}) = 0.3$
- $P(\text{open}) = P(\neg \text{open}) = 0.5$

$$P(\text{open} \mid z) = \frac{P(z \mid \text{open}) \cdot P(\text{open})}{P(z \mid \text{open}) \cdot P(\text{open}) + P(z \mid \neg \text{open}) \cdot P(\neg \text{open})}$$

$$P(\text{open} \mid z) = \frac{0.6 \cdot 0.5}{0.6 \cdot 0.5 + 0.3 \cdot 0.5} \approx 0.67$$

- z raises the probability that the door is open

Combining Evidence

- Suppose our robot obtains another observation z_2
- How can we integrate this new information?
- More generally, how can we estimate $P(x | z_1, \dots, z_n)$?

Recursive Bayesian Updating

$$P(x | z_1, \dots, z_n) = \frac{P(z_n | x, z_1, \dots, z_{n-1}) \cdot P(x | z_1, \dots, z_{n-1})}{P(z_n | z_1, \dots, z_{n-1})}$$

Markov assumption:

z_n is **independent** of $z_1, \dots, z_{n-1}$ if we know x

$$\begin{aligned} P(x | z_1, \dots, z_n) &= \frac{P(z_n | x) \cdot P(x | z_1, \dots, z_{n-1})}{P(z_n | z_1, \dots, z_{n-1})} \\ &= \eta P(z_n | x) \cdot P(x | z_1, \dots, z_{n-1}) \\ &= \eta_{1\dots n} \cdot \left[\prod_{i=1\dots n} P(z_i | x) \right] \cdot P(x) \end{aligned}$$

Example

Second Measurement

■ $P(z_2 \mid \text{open}) = 0.25$ $P(z_2 \mid \neg \text{open}) = 0.3$

■ $P(\text{open} \mid z_1) = 2/3$

$$\begin{aligned} P(\text{open} \mid z_2, z_1) &= \frac{P(z_2 \mid \text{open}) \cdot P(\text{open} \mid z_1)}{P(z_2 \mid \text{open}) \cdot P(\text{open} \mid z_1) + P(z_2 \mid \neg \text{open}) \cdot P(\neg \text{open} \mid z_1)} \\ &= \frac{\frac{1}{4} \cdot \frac{2}{3}}{\frac{1}{4} \cdot \frac{2}{3} + \frac{3}{10} \cdot \frac{1}{3}} = 0.625 \end{aligned}$$

- z_2 lowers the probability that the door is open

Actions

- Often the world is **dynamic** since
 - actions carried out by the robot
 - actions carried out by other agents
 - or just the **time** passing by change the world
- How can we **incorporate** such **actions**?

Typical Actions

- The robot **turns its wheels** to move
- The robot **uses its manipulator** to grasp an object
- Plants grow over **time** ...

- Actions are **never carried out with absolute certainty**
- In contrast to measurements, **actions generally increase the uncertainty**

Modeling Actions

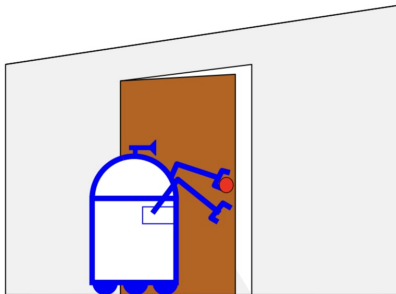
- To incorporate the outcome of an action u into the current “belief”, we use the conditional pdf

$$P(x \mid u, x')$$

- This term specifies the pdf that **executing u changes the state from x' to x**

Example

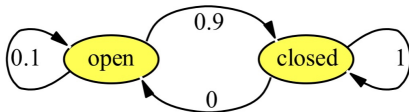
Closing the door



Example

State Transitions

$P(x \mid u, x')$ for $u = \text{"close door"}$



If the door is open, the action “close door” succeeds in 90 % of all cases

Integrating the Outcome of Actions

- Continuous case

$$p(x | u) = \int p(x | u, x') \cdot p(x' | u) dx'$$

- Discrete case

$$P(x | u) = \sum P(x | u, x') \cdot P(x' | u)$$

We will make an independence assumption to get rid of the u in the second factor in the sum

Example

The Resulting Belief

$$\begin{aligned}P(\text{closed} \mid u) &= \sum P(\text{closed} \mid u, x') \cdot P(x') \\&= P(\text{closed} \mid u, \text{open})P(\text{open}) + P(\text{closed} \mid u, \text{closed})P(\text{closed}) \\&= \frac{9}{10} \cdot \frac{5}{8} + 1 \cdot \frac{3}{8} = \frac{15}{16}\end{aligned}$$

$$\begin{aligned}P(\text{open} \mid u) &= \sum P(\text{open} \mid u, x') \cdot P(x') \\&= P(\text{open} \mid u, \text{open})P(\text{open}) + P(\text{open} \mid u, \text{closed})P(\text{closed}) \\&= \frac{1}{10} \cdot \frac{5}{8} + 0 \cdot \frac{3}{8} = \frac{1}{16}\end{aligned}$$

Bayes Filters

Framework

■ Given

- Stream of observations z and action data u

$$d_t = \{u_1, z_1, \dots, u_t, z_t\}$$

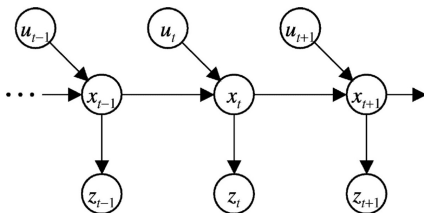
- Sensor model $P(z | x)$
- Action model $P(x | u, x')$
- Prior probability of the system state $P(x)$

■ Wanted

- Estimate of the state X of a dynamical system
- The posterior of the state is also called Belief

$$Bel(x_t) = P(x_t | u_1, z_1, \dots, u_t, z_t)$$

Markov Assumption



$$P(z_t \mid x_{0:t}, z_{1:t-1}, u_{1:t}) = P(z_t \mid x_t)$$

$$P(x_t \mid x_{1:t-1}, z_{1:t-1}, u_{1:t}) = P(x_t \mid x_{t-1}, u_t)$$

Underlying Assumptions

- Static world
- Independent noise
- Perfect model, no approximation errors

Bayes Filters

$$Bel(x_t) = P(x_t | u_1, z_1, \dots, u_t, z_t)$$

$$\text{Bayes} = \eta P(z_t | x_t, u_1, z_1, \dots, u_t) P(x_t | u_1, z_1, \dots, u_t)$$

$$\text{Markov} = \eta P(z_t | x_t) P(x_t | u_1, z_1, \dots, u_t)$$

$$\text{T. Prob.} = \eta P(z_t | x_t) \int P(x_t | u_1, z_1, \dots, u_t, x_{t-1}) P(x_{t-1} | u_1, z_1, \dots, u_t) dx_{t-1}$$

$$\text{Markov} = \eta P(z_t | x_t) \int P(x_t | u_t, x_{t-1}) P(x_{t-1} | u_1, z_1, \dots, u_t) dx_{t-1}$$

$$\begin{aligned} \text{Markov} &= \eta P(z_t | x_t) \int P(x_t | u_t, x_{t-1}) P(x_{t-1} | u_1, z_1, \dots, z_{t-1}) dx_{t-1} \\ &= \eta P(z_t | x_t) \int P(x_t | u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1} \end{aligned}$$

x – state, u – action, z – observation

Bayes Filters

$$Bel(x_t) = \eta P(z_t | x_t) \int P(x_t | u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

Algorithm bayes_filter(bel(x), d):

eta = 0

If d is a perceptual data item z **Then**

for all x do

bel'(x) = P(z|x) * bel(x)

eta = eta + bel'(x)

for all x do

bel'(x) = eta⁻¹ * bel'(x)

Else If d is an action data item u **Then**

for all x do

bel'(x) = $\int P(x|u, x') * bel(x') dx'$

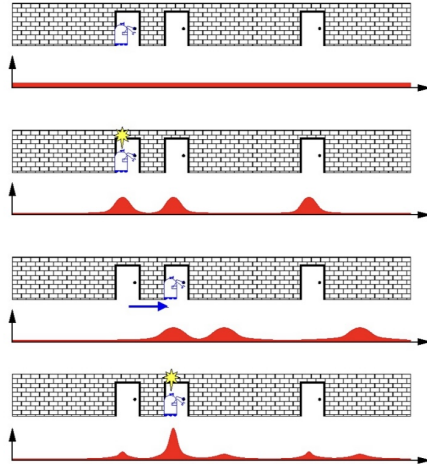
Return bel'(x)

Bayes Filters are Familiar!

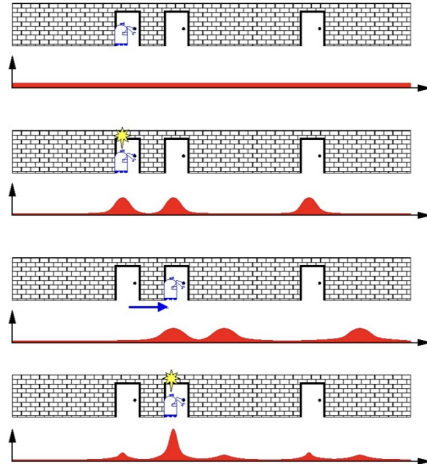
$$Bel(x_t) = \eta P(z_t | x_t) \int P(x_t | u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

- Kalman filters
- Particle filters
- Hidden Markov models
- Dynamic Bayesian networks
- Partially Observable Markov Decision Processes (POMDPs)

Probabilistic Localization



Probabilistic Localization



$$Bel(x | z, u) = \alpha p(z | x) \int_{x'} p(x | u, x') \cdot Bel(x') dx'$$

Summary

- Bayes rule allows us to compute probabilities that are hard to assess otherwise
- Under the Markov assumption, recursive Bayesian updating can be used to efficiently combine evidence
- Bayes filters are a probabilistic tool for estimating the state of dynamic systems